Combining Hilbert Feature Sequence and Lie Group Metric Space for Few-Shot Remote Sensing Scene Classification

Xiliang Chen, Guobin Zhu, and Chan Ji

***Abstract*— Few-shot learning (FSL) is a method that does not require a large number of labeled samples and has the ability to identify unseen samples. Recently, many excellent FSL methods have been developed for remote sensing scene classification (RSSC). However, for complex remote sensing scenes, the following problems have not been well addressed: 1) the order of local features and the spatial correlation between features have not been taken into consideration and 2) the current metric methods are mainly based on Euclidean space (ES), and discriminative ability is limited. Hence, we propose a framework for RSSC with FSL by combining Hilbert feature sequence and Lie group metric space (HFS-LGMS). Specifically, we adopt a lightweight four-layer convolutional neural network (CNN) as a feature extractor. For the last feature map, we regard each pixel as a multidimensional local feature. Then we organize all the features into a feature sequence along the Hilbert curve. To represent the correlation among features, we introduce the autoregressive moving average model (AMAM) to model the observed features and their spatial states. Since the parameters of the model are matrices, which are directly processed as vectors will affect performance. Therefore, we transform the matrix parameter space into the Lie group manifold space (LGMS) and introduce the Lie group intrinsic mean (LGIM) to represent the prototype. The distance between the Lie groups is measured by the geodesic distance. The similarity is measured on Lie group space. Finally, the effectiveness of HFS-LGMS is demonstrated on three public datasets.**

***Index Terms*— Autoregressive moving average model (AMAM), few-shot learning (FSL), Hilbert feature sequence and Lie group metric space (HFS-LGMS), remote sensing scene classification (RSSC).**

I. Introduction

REMOTE remote sensing scene classification (RSSC) is an important means for remote sensing image understanding. Its goal is to judge the scene category according to direct features (such as color, texture, and spectrum) or latent features of the image, so as to assist humans to understand the semantic content of remote sensing image quickly. With the development of sensor technology and image processing technology, the spatial resolution of remote sensing image is higher and higher, and the means of remote sensing image interpretation are more and more intelligent. Remote sensing scene classification (RSSC) has been widely used in various fields, including image retrieval [1], [2], disaster monitoring [3], cloud mask and cloud detection [4], [5], urban planning [6], etc.

In recent years, deep learning technology developed from computer vision has a strong feature representation ability, and the performance exceeds traditional machine learning methods, which is favored by researchers in remote sensing. Hence, RSSC methods based on deep learning, especially convolutional neural network (CNN), have become a currently mainstream research direction. Many excellent RSSC methods have been proposed, and their classification accuracy (CA) on public datasets exceeds 90% [7]. Although deep learning technology has greatly improved the performance of RSSC methods, there are also some limitations. Usually, CNN contains a large number of parameters and requires a large number of labeled samples for iterative training. If the number of samples is too small, which will lead to overfitting, the model performance will be greatly reduced. In remote sensing, a lack of labeled samples is an urgent problem [8]. Compared with natural images, the annotation of remote sensing images is much more difficult, which not only takes a lot of time but also requires annotation personnel to have rich experience and expertise [9], [10]. In addition, the previous methods are to train on the dataset with fixed scene categories. In each iteration, all the categories of training data should be input into the model from different batches. One consequence of this way is that the trained model can only identify the scene categories which are set in advance. When encountering a sample that has not been seen in the training set, this method does not work [11], [12], [13]. If you want to identify unseen samples, you need to collect datasets containing unseen samples and manually label them, and then train the model from scratch. As known, the cost of remote sensing image acquisition is huge, and it is also difficult to manually label them. The training of deep models consumes both time and computational resources. Therefore, identifying new samples by this way is almost unacceptable.

In real life, human beings have a strong ability to learn new concepts. They can summarize rules from only a few images and judge the categories of images they have never seen. Inspired by this learning mode, a method to quickly

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learn new concepts from a small number of samples was proposed, namely, few-shot learning (FSL). There are two main differences between FSL and previous learning methods. On one hand, the categories of data labels included in the training set are different. The traditional learning methods usually divide the datasets into training sets, verification sets, and test sets. The number of samples they contain is different, but the categories of labels are the same [14]. The FSL methods also divide the dataset into training set, verification set, and test set, but their label categories are independent of each other. On the other hand, training strategies are different. The previous learning method is trained based on batches. FSL is trained based on tasks. Taking training set as an example, the training set is divided into thousands of tasks. Each task contains a support set and a query set, which share a common label space. The support set contains only a small number of samples. If the support set S contains K samples, and the number of samples is N in per class, the FSL problem is called the K-way N-shot problem [15], [16], [17].

The FSL methods were first applied to natural image classification. Similar to the traditional image classification methods, the existing FSL algorithms for image classification are all developed based on deep learning. For example, data enhancement technology is used to increase the number of samples [18], image features are extracted through attention mechanism and memory mechanism [19], [20], and algorithms are designed to extract the mapping relationship between feature networks and classifiers [21], [22]. Meanwhile, migration learning [23], meta learning [24], [25], Bayesian learning [26], and graph neural network method [27], [28] are also applied to FSL tasks. In the recent two years, many RSSC methods based on FSL have been proposed [29], [30], [31], [32], [33], [34]. These methods take into account the differences between remote sensing images and natural images and greatly promote the development of few-shot RSSC. For example, Li et al. [32] proposed an adaptive matcher to alleviate intraclass difference and interclass similarity problems in remote sensing images. To reduce the interference of background for RSSC, Zeng and Geng [33] designed an attention module for feature embedding. Xing et al. [34] proposed a class dictionary sharing mechanism to address the "negative transfer" problem when knowledge learned from natural images is transferred to RSSC tasks.

Although the performance of FSL algorithms for RSSC has made great progress, these methods mainly improve the robustness of the algorithm by designing a more complex feature extractor, and there exist several problems: 1) the order of local features and the spatial correlation between features have not been taken into account and 2) the current metric methods are mainly based on Euclidean space (ES), and discriminative ability is limited. Specifically, as shown in Fig. 1, if we divide a remote sensing image which represents church into image patches, each image patch can represent local features. In Fig. 1(a), we mark the image patches containing the spire building with green dots and the dome building with red dots. In Fig. 1(b)–(d), the image patches are scanned as the order of row, column, and Hilbert curve, respectively. Fig. 1(e) shows the results of flattening the image patches according to the above three scanning sequences. It can be seen that the image patches can maintain the spatial adjacent relationship in the direction of rows or columns when the image patches are flattened along the order of rows or columns. When the image patches are flattened along the order of the Hilbert curve, the image patches can maintain the spatial adjacency relationship in the local area. Hence, we speculate that the scanning order of local features on feature map of CNN will also affect the features' spatial relationship. Zhang et al. [35] pointed out that objects in remote sensing scenes have spatial correlation. In the same way, local features on feature map of CNN also have spatial correlation. In addition, the previous FSL methods for RSSC were carried out on ES, and the Euclidean distance was used to measure the similarity of features. Xu et al. [36] pointed out that remote sensing image features represented by Lie groups have good computational performance and anti-noise ability.

Based on the above background, we propose an FSL for the RSSC framework combined with Hilbert feature sequence and Lie group metric space (HFS-LGMS). Specifically, we use a lightweight four-layer CNN as a feature extractor. For the feature map output by the last layer, we regard each pixel as a multidimensional local feature. To better maintain the spatial adjacent relationship of features, we organize all the features into a sequence along the Hilbert curve. To express the correlation between features, we introduce autoregressive moving average model (AMAM). Each feature map can establish an AMAM which is jointly represented by the observed features and space state of the features. The state of current feature is determined by the transition matrix, driving process, and the state of the previous adjacent feature. The current observation feature is determined by its state, observation matrix, and random noise. In this way, the correlation between features is established. Since the parameters of the model are matrices, the performance will be degraded if they are treated as vectors directly. Therefore, we transform the matrix parameter space into the Lie group manifold space (LGMS) and introduce Lie group intrinsic mean (LGIM) to represent the prototype on LGMS. The distance between Lie groups is measured by the geodesic distance. Except that the parameters of the four-layer CNN need to be trained, we have not added additional parameters that need to be trained in HFS-LGMS. In addition, our method does not require fine-tuning during the test stage. Our contributions can be summarized as follows.

1) Considering the influence of the arrangement order of high-level features of remote sensing images for RSSC with FSL methods, we flatten local features along the Hilbert curve to better maintain the spatial adjacent relationship of remote sensing image features.
2) We introduce the AMAM to model the observation features and the space state of the features and describe the spatial correlation between local features of remote sensing images.
3) We propose a more robust feature representation method. The parameters of the AMAM are embedded into LGMS, and the prototype of the features on Lie group

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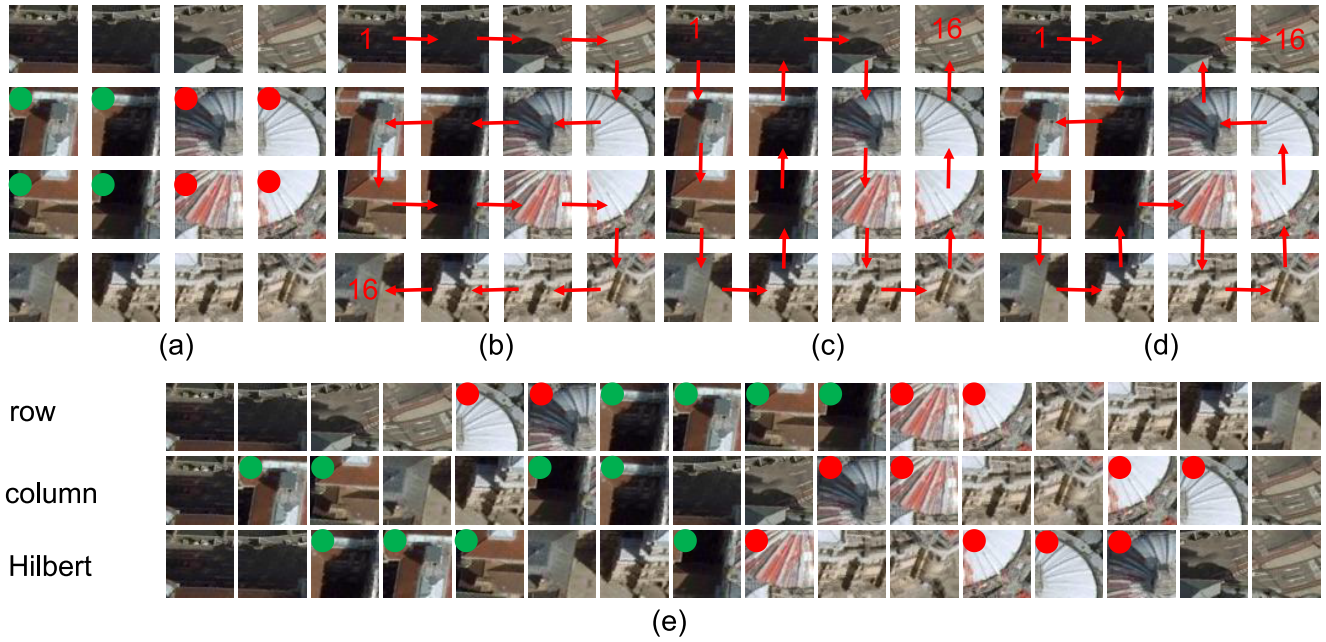

Fig. 1. Influence of different scanning sequences on spatial relationships of local image patches. (a) Special image patches' marking. (b) Scanning by row sequence. (c) Scanning by column sequence. (d) Scanning by Hilbert sequence. (e) Flattening the local image patches according to different scanning sequences.

space is defined by LGIM. The distance between two features is measured by the geodesic distance.

The rest of this article is organized as follows: In Section II, we will introduce the work related to RSSC, RSSC with FSL and the application of Lie group in image classification. In Section III, we will introduce the detailed methodology involved in the HFS-LGMS framework. The experimental results on several public datasets are presented in Section IV. Finally, in Section V, we will summarize the content of this article and draw a conclusion.

II. Related Work

A. Remote Sensing Scene Classification

RSSC based on deep learning is a hot research direction at present, and a series of excellent methods have been proposed in recent years. For example, Zhao et al. [37] proposed a self-distillation network, which is composed of multiple branches. Through knowledge distillation, knowledge can be condensed to the main branch. Other branches are discarded during the inference process, which greatly reduces the number of parameters and improves the prediction efficiency. Chen et al. [38] developed an algorithm that can simultaneously extract multiscale features and global features of remote sensing images. To distinguish easily confused images, Zhang et al. [39] selected two most similar images to form an image pair and expressed their features through self-representation and mutual representation, so as to further mine the nuances of image pairs to discriminate them. Although these methods have excellent performance, they require a large number of labeled samples, so researchers try to explore methods that can alleviate the difficulty of obtaining labeled samples in remote sensing. Currently popular methods that can reduce the demand for labeled data include self-supervised learning [40], [41], [42] and transfer learning [43], [44]. Chen et al. [40] proposed a self-supervised learning network with an encoder–decoder module. which can extract supervised signals from unlabeled remote sensing images, and then fine-tune them on labeled samples according to a specific task. Kang et al. [41] and Stojnić and Risojević [42] also completed self-supervised learning by constructing an auxiliary task using unlabeled data, and then applied the trained feature extractor to the RSSC task. Transfer-learning-based methods usually use rich labeled natural datasets to train the model and then fine-tune it on remote sensing datasets. Then, these methods still cannot solve the problem of identifying scene categories with only a few labeled samples, nor can they identify sample categories that have never been seen before. The HFS-LGMS framework we proposed has this ability, which is the fundamental difference between our method and the above methods.

B. FSL for RSSC

In remote sensing, the scarcity of labeled samples hinders the further development of remote sensing applications. The RSSC methods based on deep learning need to input a large number of labeled samples to train a stable model. Such methods are ineffective when the number of samples is extremely limited and scene categories that have never been seen need to be identified. In this context, the method of RSSC with FSL emerged as the times require and achieved good results [29], [31]. For example, Li et al. [32] designed an attention module to guide the model to focus on specific areas, and then combined the output of the attention module with the feature map of the network to enhance the discrimination. Cheng et al. [45] improved the classical prototype network (PN) [46] and used the supervision information of the sample and the confidence score of the query sample to generate the prototype. In addition, they further corrected the prototype

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according to the interactive information between them to make the prototype more representative. Zeng and Geng [33] and Li et al. [47] used contrast learning as a specific task and contrast loss function as a constraint condition, so as to reduce the interclass differences in sample features, increase the interclass distance, and improve the discriminant ability of the RSSC algorithm with FSL. Cui et al. [48] designed a meta-kernel network to map the indistinguishable low-dimensional features extracted from the network to the high-dimensional space through the meta-kernel strategy, which makes the boundaries of each category more obvious, greatly reduces the variance ratio of intraclass and interclass, and improves the accuracy of RSSC with FSL. Chao and Li [49] first proposed data quality assessment method for RSSC tasks with FSL and adopted KNN distance entropy as the evaluation index for the sample quality. The author believes that in the case of a limited number of samples, it is helpful to improve the classification effect through selecting high-quality samples. The above methods represent features in ES without considering the order of features and the spatial correlation between features. Our method takes these factors into account to construct a more robust RSSC framework with FSL.

C. Zero-Shot Learning (ZSL) for RSSC

ZSL is another way to identify new scenes that have never appeared in the training set. ZSL methods can infer and identify the samples with unseen classes by learning the knowledge from the samples with seen classes and combining the auxiliary information (e.g., attributes). In general, the samples with seen classes are used for training, and the samples with unseen classes are used for testing. Different from FSL, ZSL needs to introduce some additional information, which are usually the descriptions of all the classes. The research about ZSL can be traced back to 2008. Larochelle et al. [50] proposed a method for character classification in the case of mutual exclusion of testing classes and training classes. Lampert et al. [51] extended this concept to the field of computer vision. Since then, ZSL has attracted more and more researchers' attention [52], [53]. In recent years, the development of deep learning technology has further promoted the development of ZSL and has been widely used in image classification [54], [55], [56]. Li et al. [57] first applied ZSL to RSSC. They used the visual similarity between images with the same scene class and proposed a label thinning method based on sparse learning to suppress the noise in the zero-shot classification results. Li et al. [58] proposed a new deep alignment network with a series of well-designed optimization constraints, which can simultaneously solve the problem of ZSL and generalized ZSL for RSSC. Li et al. [59] proposed a deep cross-modal embedding network based on local protection for RSSC. ZSL has developed rapidly in natural image classification, but has developed slowly in RSSC. On the other hand, the universal pretrained natural language models are usually used to extract semantic vectors based on category names to form semantic features of different categories, but the universal natural language models often fail to accurately describe the category's features of remote sensing scenes. Therefore, the ZSL methods developed in natural image recognition are difficult to achieve satisfactory results when they are directly applied to RSSC. In addition, since there are no samples with seen classes in the test stage, only description information of classes can be used. Hence, ZSL is much more complex than FSL.

D. Image Classification on LGMS

Unlike ES on the plane, Lie groups are in manifold space. The features represented on LGMS have good computational performance and strong robustness [36]. In recent years, Lie group features have been widely used in image classification. For example, Xu et al. [36] used LGIM to represent the global features of samples and used geodesics to measure the distance between the Lie group features to construct a lightweight and discriminative remote sensing scene classifier. Xu et al. [60] simultaneously extracted local features and global features in Lie group space from remote sensing image, and then combined them with CNN to guide the model's attention through regional influence network and achieved excellent performance. Many other methods are based on the regional covariance matrix. The covariance matrix is located in the LGMS, which has the advantage of low dimensions and can fuse multiple features at the same time. Therefore, it is also widely used. For example, He et al. [61] combined skip connection and covariance pool to construct a skip connection covariance (SCCov) network for RSSC. This method can not only integrate multiresolution feature maps but also make full use of the second-order information contained in the feature map, so that the learned features are more representative. He et al. [62] proposed a multilayer stacked covariance pool (MSCP) method for RSSC. They stacked feature maps of different layers together and then calculated the covariance matrices of features in different local regions. This method naturally combines multilayer feature maps and achieves better performance than single-layer covariance pooling models. Li et al. [63] applied covariance pooling to FSL for the first time. They proposed covariance representation and covariance metric based on distribution consistency and proved the superiority of this method. Our method also involves Lie group features. Different from the above methods, we do not directly use the feature vectors to construct the Lie group, but embed the parameters of the AMAM into the Lie group space. These parameters contain the latent sequence information and spatial correlation of local features, which are more representative.

III. Proposed Method

A. Hilbert Curve

The German mathematician David Hilbert proposed the Hilbert curve [64] in 1891. This curve can continuously traverse all the points in the square and fill the entire space, so it is also called the Hilbert space-filling curve. The space-filling curve is usually used to transform the multidimensional space into a 1-D space. It connects the elements in each cell in the multidimensional space (each element is regarded as a point in the multidimensional space) through a group of oriented line segments to form a 1-D element sequence, and the elements in each cell are traversed once. Spatial-filling curves are widely used in tasks such as data compression,

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spatial clustering, and image processing [65]. The value of Hilbert curve in both science and art is derived from its three characteristics: first, it can fill the entire plane space; second, it is very tortuous and continuous but not derivable; finally, it has self-similarity. Moon et al. [66] believed that when using 1-D space to represent data on multidimensional space, the Hilbert curve has obvious advantages over other data compression schemes based on the space-filling curve and can well preserve the spatial locality of multidimensional data. Voorhies [67] proved that the Hilbert curve has good coherence between adjacent image patches. Considering that the Hilbert curve can maintain the spatial locality of the image patches, Bappy et al. [68] divided the image into image patches and resampled the features of them, and then flattened the features of the image patches into the LSTM network according to the order of the Hilbert curve, which greatly enhanced the ability of LSTM to capture the correlation information between local features.

The Hilbert curve is the limit of a series of fractal curves H_N, where N denotes the order of the Hilbert curve and N is a positive integer. We can consider H_N as a curve covering a $2N \times 2N$ square matrix. The construction method of Hilbert curve is very simple. When $N = 1$, a square is divided into four equal small squares, and then from the small square in the upper left corner to the small square in the upper right corner, the center of the small square is connected by line segments in turn, and the construction of the first-order Hilbert curve is completed. H_N $(N > 1)$ can be constructed by the following steps: 1) rotate H_{N-1} counterclockwise by 90° and scale it as half size and place it in the upper left corner; 2) rotate H_{N-1} clockwise by 90° and scale it as half size and place it in the upper right corner; 3) scale two H_{N-1} as half size and place them in the lower left and lower right corners, respectively; and 4) three line segments are used to connect 1), 2), and 3), and the length of these line segments is the same as the length of each side of the small square. The curve obtained when the number of iterations is N is called the N-order Hilbert curve. Fig. 2 shows the geometrical shape of the N-order Hilbert curve, $1 \leq N \leq 3$. The dotted line represents the square connected by the Hilbert curve, the solid line represents the Hilbert curve, and the red highlighted line segments represent the connections between the endpoints of the fractal curve. The results show that the adjacent squares in 2-D space are still in adjacent positions in the 1-D space, which minimizes the damage to the original data structure.

In this article, we use the second-order Hilbert curve to flatten the remote sensing image feature map. The size of the feature map we output is 4×4 pixels, and each pixel represents a multidimensional feature point. According to the construction method of the second-order Hilbert curve, all the feature points can be connected in order and mapped into a 1-D space.

B. Modeling From Feature Sequence

For each remote sensing image, the feature sequence contains its local features and relative spatial position information of the features, but it cannot reflect the correlation between the features. Xu et al. [69] cut the image into image patches and used feature vector of each image patch to represent the local features of the image. Considering the spatial correlation between different image patches, the author introduced the AMAM to model the features of image patches sequence, and then further mapped the model parameters to Grassmann manifold for image classification. Inspired by this work, we also introduced AMAM to model feature sequences. Different from this work, we do not directly model the original features of the image, but encode more representative high-level features output by CNN in accordance with the order of the Hilbert curve, and then use AMAM to construct a model to represent the correlation between features.

1) AMAM: For a specific local feature sequence, each local feature is obviously not independent of each other, but there is a certain spatial correlation. AMAM can establish the relationship between different local features. We regard each feature vector as the output of AMAM and express the dependencies between features through a series of intermediate parameters. For a feature map F with the size of $m \times h \times w$, we flatten it into a feature sequence along the Hilbert curve. The number of features is $h \times w$ and the dimension is m. That is, $\{f_i\}_{i=1,2,\ldots,n}, n = h \times w$. The local feature $f_i \in R^m$ with sequence can be fit by a linear function based on the current state and observation noise. The spatial sequence relationship between features can be represented by the transition relation between states $z_i \in R^n$. Therefore, each feature map can be represented by the following AMAM:

$$\begin{cases} f_i = S \times z_i + \phi_i, & i \in \{1, 2, \ldots, n\} \\ z_{i+1} = M \times z_i + \varphi_i, & i \in \{1, 2, \ldots, n-1\} \end{cases} \tag{1}$$

where $S \in R^{m \times n}$ denotes the observation matrix, and $M \in R^{n \times n}$ denotes the transition matrix. $z_1 \in R^n$ denotes the initial state vector. $\{f_i\}_{i=1,2,\ldots,n}$ represents the local feature sequence of the feature map, and $\{z_i\}_{i=1,2,\ldots,n}$ represents the spatial state sequence corresponding to local features. ϕ_i denotes the observation noise and φ_i denotes the state noise, and they conform to the normal distribution, $\phi_i \sim N(0, Q)$ and $\varphi_i \sim N(0, P)$. From the model, we can see that the state of the $i+1$ local feature is determined by the state of the i local feature, transition matrix M, and state noise φ_i. The observed value of the current local feature is determined by the state of the current local feature, observation matrix S, and observation noise ϕ_i.

The relationship between local features and state transitions is shown in Fig. 3. The CNN outputs a feature map with the size of $64 \times 4 \times 4$. We flatten it into a feature sequence with a length of 16 according to the direction of the Hilbert curve. The dimension of the feature is 64. Each feature can be represented as a linear function of its corresponding spatial state and observation noise.

2) Parameter Estimation: For a given local feature sequence $\{f_1, f_2, \ldots, f_n\}$, according to (1), parameters $S, M, Q, and P$ are unknown and need to be estimated. The parameters of AMAM can be estimated by the maximum likelihood method

$$S, M, Q, P = \arg \max_{S,M,Q,P} p(f_1, f_2, \ldots, f_n). \tag{2}$$

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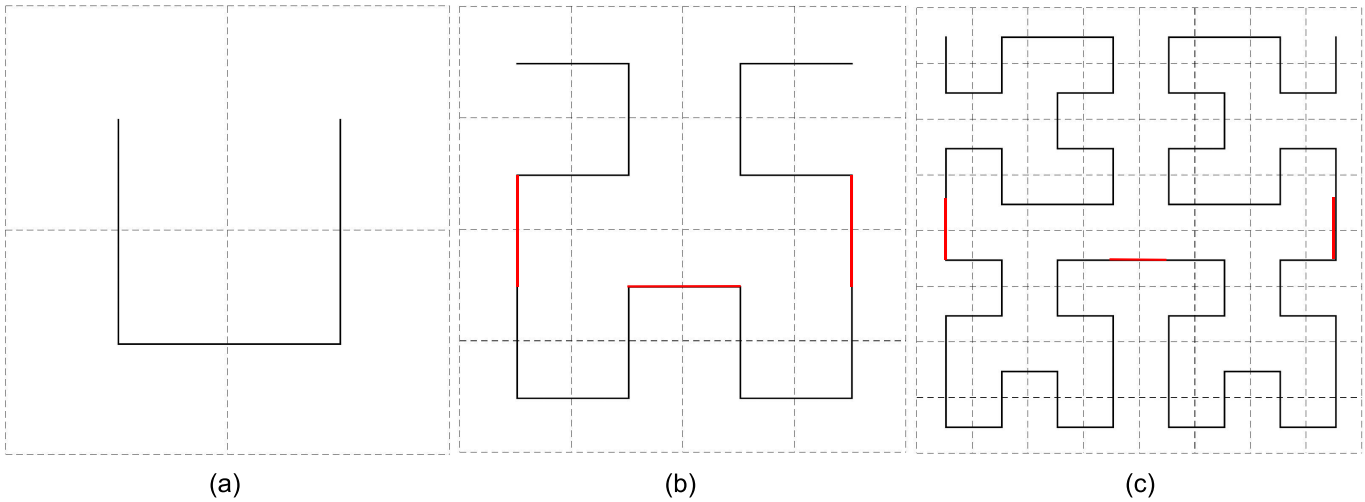

Fig. 2. Hilbert curves with different orders. (a) First-order. (b) Second-order. (c) Third-order. The red highlighted line segments represent the connections between the endpoints of the fractal curve.

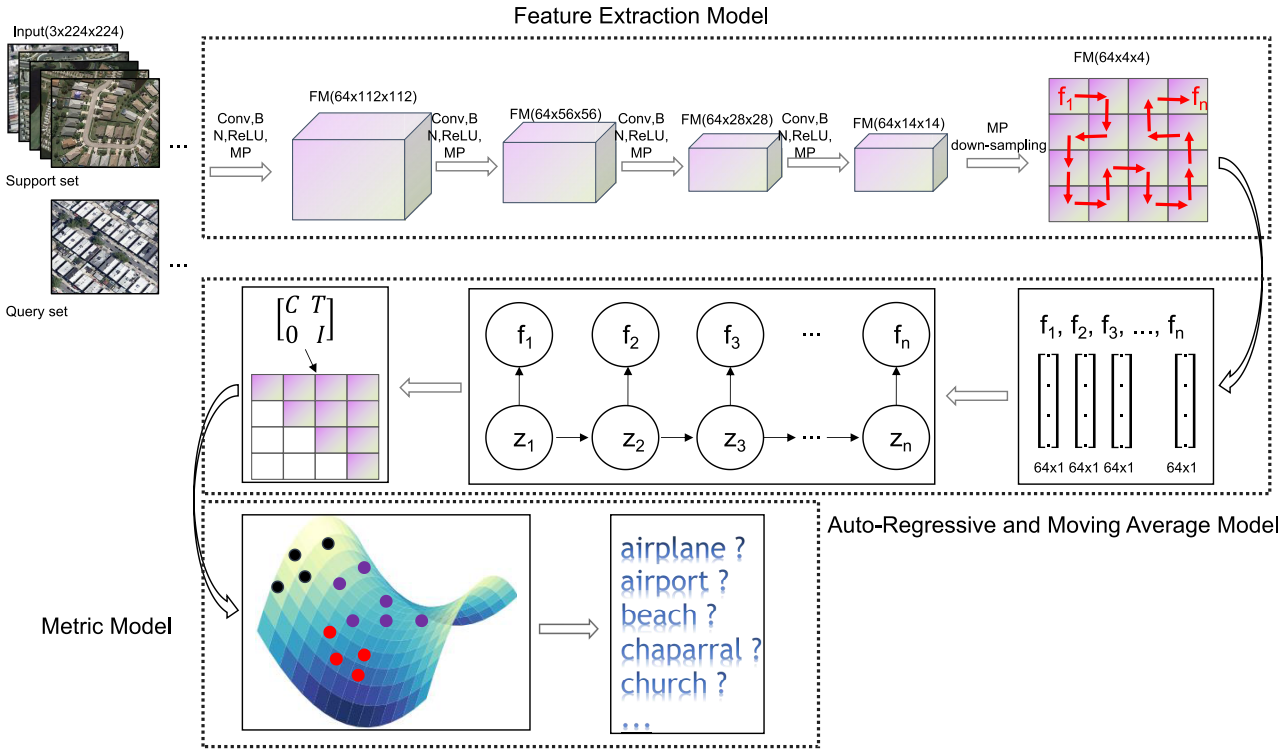

Fig. 3. Overall architecture of HFS-LGMS (Conv denotes convolution, BN denotes batch norm, MP denotes max pool, and FM denotes feature map).

Generally, these parameters can be solved by the maximum likelihood method. However, the local feature vector dimension of images is usually high, and it is inefficient to learn these parameters by the maximum likelihood method. Hence, approximate solution [70] is adopted to estimate model parameters.

For a given feature map, $F = [f_1, f_2, \ldots, f_n] \in R^{m\times n}$ represents the feature matrix composed of feature sequence. $Z = [z_1, z_2, \ldots, z_n] \in R^{n\times n}$ represents the spatial state matrix corresponding to the feature sequence. We perform singular value decomposition for F. $F = U\Sigma V^T$, $U \in R^{m\times n}$, $U^T U = I$, $V \in R^{n\times n}$, and $V^T V = I$. The parameters are estimated as follows:

$$\begin{cases} \hat{S} = U \\ \hat{Z}_1^n = \Sigma V^T \\ \hat{M} = \hat{Z}_2^n \left(\hat{Z}_1^{n-1}\right)^+ \\ \hat{Q} = \dfrac{1}{n-1}\displaystyle\sum_{i=1}^{n-1} \hat{\mu}_i \hat{\mu}_i^T \\ \hat{P} = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} \hat{v}_i \hat{v}_i^T \end{cases} \tag{3}$$

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where $\hat{Z}_1^n = [\hat{z}_1, \hat{z}_2, \ldots, \hat{z}_n]$, $\hat{Z}_1^{n-1} = [\hat{z}_1, \hat{z}_2, \ldots, \hat{z}_{n-1}]$, and $\hat{Z}_2^n = [\hat{z}_2, \hat{z}_3, \ldots, \hat{z}_n]$. They are all hidden state estimation matrices. $(\hat{Z}_1^{n-1})^+$ is the pseudoinverse matrix of $\hat{Z}_1^{n-1}$, $\hat{\mu}_i = \hat{z}_{i+1} - \hat{M} \times \hat{z}_i$, where $i \in \{1, 2, \ldots, n-1\}$. $\hat{v}_i = f_i - \hat{S} \times \hat{z}_i$, where $i \in \{1, 2, \ldots, n\}$.

C. Metric on LGMS

In this section, we construct a Lie group manifold structure according to the parameters of AMAM, and then use the distance metric of LGMS to define the prototype, which is finally used for RSSC with FSL.

1) Constructing Lie Group From Parameters of AMAM: We use the parameters of AMAM to construct a matrix with the upper triangular structure. The upper triangular matrix L is defined as follows:

$$L = \begin{bmatrix} C & T \\ 0 & I_{n-1} \end{bmatrix} \tag{4}$$

where $T = [\hat{M}z_1, \hat{M}z_2, \ldots, \hat{M}z_{n-1}] \in R^{n\times(n-1)}$, and $\hat{M}$ denotes the transition matrix. $z_1, z_2, \ldots, z_{n-1}$ denotes the state vectors of the local features. I_{n-1} denotes the identity matrix. We perform Cholesky decomposition for the positive definite symmetric matrix $\hat{Q}$ in (3), i.e., $\hat{Q} = C^T C$.

We will demonstrate that all the matrices L form a Lie group G according to the definition of Lie group. We assume that L_1 and L_2 are two upper triangular matrices satisfying the structure in (4), then the multiplication between them can be expressed as

$$L_1 L_2 = \begin{bmatrix} C_1 & T_1 \\ 0 & I_{n-1} \end{bmatrix} \begin{bmatrix} C_2 & T_2 \\ 0 & I_{n-1} \end{bmatrix} = \begin{bmatrix} C_1 C_2 & C_1 T_2 + T_1 \\ 0 & I_{n-1} \end{bmatrix}. \tag{5}$$

Since C_1 and C_2 are the upper triangular matrices, the multiplication result of C_1 and C_2 is also an upper triangular matrix. Hence, the multiplication of group elements L_1 and L_2 is also an upper triangular matrix. The inversion operation of any group element can be expressed as

$$L^{-1} = \begin{bmatrix} C & T \\ 0 & I_{n-1} \end{bmatrix}^{-1} = \begin{bmatrix} C^{-1} & -C^{-1}T \\ 0 & I_{n-1} \end{bmatrix}. \tag{6}$$

Since the inverse matrix C^{-1} of the upper triangular matrix C is still an upper triangular matrix, the inverse of the group element L is also an upper triangular matrix. In summary, the results of the multiplication operation of any two group elements and the inverse operation of any group element are consistent with the upper triangular structure defined in (4). Therefore, we can consider that all the upper triangular matrices L form Lie groups.

2) Definition of Prototype on LGMS: Snell et al. [46] proposed a PN for FSL, and they defined the average of the feature vectors in the support set for each category of sample as a prototype. This way is calculated directly based on the Euclidean distance. Hence, this prototype is only applicable to ES. In our method, all the features are on the LGMS, and the Euclidean distance cannot represent the distance between two points on the manifold space [36]. So we need to define a prototype on LGMS. On LGMS, the distance between any two elements is represented by the geodesic distance

$$d(L_1, L_2) = \left\|\log\left(L_1^{-1}, L_2\right)\right\| \tag{7}$$

where $||\cdot||$ denotes the Frobenius norm. We introduce LGIM to define the prototype on LGMS. We assume that $\bar{L}$ represents the LGIM of a series of points $\{L_1, L_2, \ldots, L_n\}$ on the LGMS, and $\bar{L}$ is defined as

$$\bar{L} = \arg\min_{L\in G} \sum_{i=1}^{n} \| \log(L^{-1} L_i) \|^2 \tag{8}$$

where $L \in G$ denotes that L belongs to LGMS. $\bar{L}$ satisfies two conditions: $\bar{L} \in G$, and the sum of squares of the geodesic distance between $\bar{L}$ and each point in $\{L_1, L_2, \ldots, L_n\}$ is minimal. Sung et al. [77] proposed a method for estimating $\bar{L}$ based on gradient descent, and an optimization function was designed

$$f(L) = \frac{1}{2n} \sum_{i=1}^{n} d(L, L_i)^2. \tag{9}$$

The gradient of function f

$$\nabla f(L) = -\frac{1}{n} \sum_{i=1}^{n} \log(L^{-1} L_i). \tag{10}$$

Given the estimated value μ_m of the LGIM in the kth iteration, the estimated value in the $k+1$th iteration is as follows:

$$\mu_{m+1} = \exp\left(-\frac{\sigma}{n} \sum_{i=1}^{n} \log\left(\mu_m^{-1} L_i\right) \right) \tag{11}$$

where σ denotes the step size in the gradient descent algorithm. The step size determines the length of each step along the negative direction of the gradient in the process of gradient descent iteration; when the value of σ is too large, it may cross the extreme point, while if the value is too small, the convergence speed will be too slow. Hence, we will investigate the optimal value of σ in Section IV-F. When $m = 1$, $m_1 = L_1$. There is a detailed calculation process in [36] and [71]. If the final value obtained through iteration is $\hat{\mu}$, then $\bar{L} = \hat{\mu}$, and $\bar{L}$ is the LGIM. Referring to the definition of prototype on ES, we define the LGIM as the prototype on LGMS.

D. Overall Framework of HFS-LGMS

Fig. 3 shows the framework of HFS-LGMS, which consists of a feature extraction model (FEM), an AMAM, and a metric model. The FEM is mainly composed of four CNN layers, and each layer contains a 3×3 convolution (Conv), a batch normalization (BN), a rectified linear unit (ReLU) activation nonlinearity layer, and a 2×2 max-pooling (MP) layer. The number of channels outputted by each convolutional layer is 64. Supposing that we input an image with the size of $3 \times 224 \times 224$, the dimensions of the four convolutional layer feature maps are $64 \times 112 \times 112$, $64 \times 56 \times 56$, $64 \times 28 \times 28$, and $64 \times 14 \times 14$, respectively. In the end of CNN, we convert the dimension of the feature map into

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TABLE I
SPLITTING CRITERIA FOR THE DATASETS

Name	Training Categories	Validation Categories	Testing Categories
NWPU dataset	Wetland, Stadium, Sparse residential, Ship, Sea ice, Roundabout, Rectangular farmland, Railway, Palace, Mountain, Mobile home park, Meadow, Island, Lake, Harbor, Golf course, Freeway, Desert, Cloud, Church, Chaparral, Bridge, Beach, Baseball diamond, Airplane	Thermal power station, Terrace, Tennis court, Storage tank, Snowberg, Runway, Railway station, Overpass, Industrial area, Commercial area	River, Parking lot, Medium residential, Intersection, Ground track field, Forest, Dense residential, Circular farmland, Basketball court, Airport
AID dataset	Forest, School, Medium Residential, Port, Pond, Mountain, Resort, Parking, Bare Land, Farm land, Bridge, Commercial, Baseball Field, Church, Bridge, Industrial	Stadium, Beach, Square, Meadow, Playground, Center, Park	Sparse Residential, Viaduct, Airport, Storage Tanks, River, Railway Station, Desert
UCM dataset	Parking lot, Overpass, Medium residential, Harbor, Freeway, Dense residential, Chaparral, Buildings, Baseball diamond, Agricultural	Storage tanks, Runway, Intersection, Forest, Airplane	Tennis court, River, sparse residential, Mobile home park, Golf course, Beach

$64 \times 4 \times 4$ by downsampling. Then we flatten the feature map as the order of the Hilbert curve. The dimension of each local feature is 64×1, and we use AMAM to construct a model to represent the correlation between features. Finally, we map the parameters of AMAM to the Lie group space, introduce LGMS to define prototype, and use the geodesic distance to measure the distance between Lie group elements. In our method, only the parameters in the FEM need to be trained in the training stage. In the test stage, all the parameters do not need to be trained.

IV. EXPERIMENTS

A. Datasets' Description

To evaluate the performance of our proposed method for RSSC with FSL, we conducted experiments on three commonly used public datasets, which include NWPU-RESISC45 (NWPU) [72], aerial image data (AID) [73], and UC-Merced Land Use (UCM) [74] datasets. The total image number of the NWPU dataset is 31 500, the image size is 256×256, which contains 45 categories, there are 700 images per category, and the spatial resolution of the dataset is from 0.2 to 30 m. The AID dataset contains 10 000 samples with the size of 600×600. There are 30 scene categories in the dataset, each category contains about 220–420 images, and the spatial resolution of the AID dataset is from 0.5 to 8 m. The UCM dataset contains 2100 images with 21 categories, the image size is 256×256, there are 100 images in each category, and the spatial resolution is about 0.3 m. To make a fair comparison with other FSL methods, we follow the standard splits to produce training set, validation set, and testing set. NWPU set is the most challenging dataset and contains the most abundant scene categories. We assign 25, ten, and ten categories for training, validation, and testing sets, respectively. For the AID dataset, 16 categories are selected as the training set, and both the validation and testing sets contain seven categories. For the UCM dataset, we divide all the categories into training set, validation set, and testing set according to the ratio of 10:5:6. The scene categories contained in each dataset are detailedly reported in Table I.

B. Experimental Setup and Hyperparameter Settings

In our experiment, we uniformly resized all the remote sensing images to 224×224. We adopted an episode-based strategy for training. The datasets are divided into training sets D_{tr}, validation sets D_{va}, and test sets D_{te}, but their label types are independent and disjoint, i.e., $D_{\text{tr}} \cap D_{\text{va}} = \emptyset$, $D_{\text{tr}} \cap D_{\text{te}} = \emptyset$, and $D_{\text{te}} \cap D_{\text{va}} = \emptyset$. The datasets are divided in FSL based on tasks. Taking the training set as an example, the training set is divided into thousands of tasks. Each task contains a support set and a query set, which share a common label space. The support set contains five different categories of samples, and the number of samples for each category is one or five. The categories of the query set sample are consistent with the support set, but the query set has no limitation on the number of samples. Here we randomly select 15 samples for each category. In the data preprocessing stage, we randomly generate 200 episodes with the training set, 500 episodes with the validation set, and 2000 episodes with the testing set to prepare for FSL. We use the average CA or CAs of 2000 tasks on the testing set with 95% confidence as the evaluation metric of the model performance. We used the SGD optimizer to optimize the model, and the learning rate is set to $1e^{-3}$. Through many experiments, σ is set 0.3. The PyTorch framework is selected to implement our model. All the experiments are performed on a computer equipped with the NVIDIA RTX 2080Ti GPU.

C. Comparison With Other FSL Methods

To explore the performance of our proposed method (HFS-LGM) for RSSC with FSL, we implement the experiments

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TABLE II
CAs (%) OF FSL FOR RSSC ON THE NWPU DATASET

Method	Five-way	
	One-shot	Five-shot
MN [75]	42.81±0.12	47.71±0.26
Meta-SGD [76]	60.56±0.28	74.33±0.65
TPN [81]	66.51±0.87	78.50± 0.56
RN [77]	66.05±0.32	78.24±0.47
PCFGNet [78]	72.05 ±0.75	**85.07 ± 0.45**
IDLN [8]	**75.25± 0.75**	84.67 ± 0.23
PN [46]	41.57±0.42	63.09±0.08
CovaMNet [63]	43.62±0.67	67.23±0.15
DLA-MatchNet [32]	68.80±0.70	81.63±0.46
GES-Net [79]	70.83± 0.85	82.27± 0.55
MAML [25]	47.32±0.10	63.03±0.55
MKN [48]	65.84 ±0.89	82.67± 0.55
SCL-MLNet [47]	62.21±1.12	80.86±0.76
LLSR [30]	51.43	72.90
TAE-Net [80]	69.13± 0.83	82.37±0.52
RS-MetaNet [29]	46.21±0.58	68.75±0.70
TSC [33]	73.26±0.15	84.62±0.35
SPNet [45]	67.84±0.87	83.94 ±0.50
HFS-LGMS (ours)	**76.81±0.18**	**87.14±0.35**

TABLE III
CAs (%) OF FSL FOR RSSC ON THE AID DATASET

Method	Five-way	
	One-shot	Five-shot
Meta-SGD [76]	53.14±1.46	66.94±1.20
RN [77]	55.26±1.14	72.81±0.93
PN [46]	54.79±0.60	69.36±0.10
CovaMNet [63]	56.39±0.34	72.95±0.6
DLA-MatchNet [32]	57.21±0.82	73.45±0.61
MAML [25]	47.98±0.72	61.72±0.81
SCL-MLNet [47]	59.46±0.96	**76.31±0.68**
RS-MetaNet [29]	53.37± 0.56	72.59±0.73
SIB [82]	60.72± 0.78	71.86 ±0.49
CAN + T [83]	**63.82± 0.56**	73.77±0.41
HFS-LGMS (ours)	**65.96±1.15**	**77.86±0.68**

on three public datasets which include NWPU, AID, and UCM. We compare the experimental results with many other excellent FSL methods. Many classical methods (e.g., MN [75], Meta-SGD [76], RN [77], PN [46], and MAML [25]) come from the field of computer vision which are widely used in natural images' classification. In addition, for the application in RSSC, researchers have developed many FSL methods (e.g., PCFGNet [78], IDLN [8], DLA-MatchNet [32], GES-Net [79], MKN [48], LLSR [30], TAE-Net [80], TSC [33], and SPNet [45]) with better performance. However, some of these methods (e.g., Meta-SGD [76], LLSR [30], and MAML [25]) also require fine-tuning during the test stage, which makes them very inconvenient. Like most metric-based FSL methods, our method can be used directly during the test stage without any additional fine-tuning. Tables II–IV report the comparison results of our method with other methods. To make the comparison more prominent, we display the best and second best results with bold text.

Table II shows the CAs on the NWPU dataset; we compared the experimental results with 17 other methods. We can see that there are relatively poor effects when the methods in computer vision were directly applied to the RSSC. For example, among these methods coming from the field of computer vision, PN [46] achieves the highest CA. In the five-way one-shot group, the CA is 41.57%. The CA is 63.09% when the group is five-way five-shot. The CAs are lower than all the methods specifically developed for RSSC. What is more important, the performance our method surpasses all other methods. In the five-way one-shot group and five-way five-shot group, the CAs of HFS-LGMS are 76.81% and 87.14%. However, when in the case of five-way one-shot, the suboptimum method is IDLN [8] with the CA of 75.25%; the CA is 1.56% lower than our method. When in the case of five-way five-shot, the suboptimum method is PCFGNet [78] with the CA of 85.07%; the CA is 2.07% lower than our method. Compared with other datasets, NWPU contains the largest number and most complex classes of scene. Hence, the experimental results indicate that our method can maintain best performance on complex RSSC datasets.

TABLE IV
CAs (%) OF FSL FOR RSSC ON THE UCM DATASET

Method	Five-way	
	One-shot	Five-shot
MN [75]	34.22±0.77	53.12±0.20
Meta-SGD [76]	51.78±1.05	65.47±0.65
TPN [81]	53.36±0.77	68.23± 0.52
RN [77]	48.60±0.73	62.34±0.66
PN [46]	52.36±0.21	68.97±0.72
CovaMNet [63]	53.76±1.25	71.18±0.39
DLA-MatchNet [32]	53.76±0.62	63.01±0.51
MAML [25]	49.03±0.28	60.89±0.27
MKN [48]	58.45 ±0.54	**77.92± 0.32**
SCL-MLNet [47]	51.37±0.79	68.09±0.92
LLSR [30]	39.47	57.4
TAE-Net [80]	**60.21± 0.72**	77.44±0.51
RS-MetaNet [29]	52.31±1.12	67.21±0.44
TSC [33]	55.11±0.68	69.20±0.64
SPNet [45]	57.64±0.73	73.52 ±0.51
HFS-LGMS (ours)	**63.05±0.78**	**79.68±0.81**

The experimental results on the AID dataset are displayed in Table III. It can be seen that our method still has highest CAs. In the five-way one-shot group, the CA of HFS-LGMS is 65.96%, the CA of suboptimum method (CAN + T [83]) is 63.82%, and our CA is 2.14% than the suboptimum method. In the case of five-way five-shot, the second best method is SCL-MLNet [47], and the CA is 76.31%. However, the CA of our method is 77.86%. It is obvious that the CA of our method is improved by 1.55%. It is worth noting that SCL-MLNet contains six CNN layers, and our method only contains four CNN layers. Hence, compared with SCL-MLNet, our method not only has a better performance but also has fewer training parameters.

In addition, we have conducted experiments on the UCM dataset, Table IV shows the experimental results. We can see that the CAs of our proposed method reach 63.05% and 79.68%, respectively, in the five-way one-shot group and five-way five-shot group. For the former, the suboptimum method is TAE-Net [80], in which CA is 60.21%; the CA is 2.84% lower than our method. For the latter, the suboptimum method is MKN [48], in which CA is 77.92%, and the CA of our method is improved by 1.76%.

Through analysis, we can see that our method outperforms other methods on three public datasets for RSSC tasks. Compared with previous methods, our innovation is that we consider not only the order of features but also the

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TABLE V
FIVE-WAY FSL CAS (%) FOR RSSC ON THREE DATASETS WITH DIFFERENT COMPONENTS

Settings	NWPU		AID		UCM	
	One-shot	Five-shot	One-shot	Five-shot	One-shot	Five-shot
FEM (ES)	74.26±0.62	83.96±0.11	63.89±0.19	75.01±0.38	60.57±0.26	76.94±0.64
FEM (LGMS)	75.33±0.36	84.67±0.85	64.48±0.43	75.54±0.32	61.26±0.45	77.62±0.43
FEM+AMAM	76.14±0.59	85.91±0.41	65.14±1.05	76.91±0.11	62.35±0.74	78.5±0.3
FEM+AMAM+LGIM (ours)	**76.81±0.18**	**87.14±0.35**	**65.96±0.15**	**77.86±0.68**	**63.05±0.78**	**79.68±0.81**

spatial correlation between features. We flatten the local features along the Hilbert curve, which maintain the spatial neighborhood relationship of the features. Then, we use AMAM to model the spatial relationship between local features, and then map the parameters of the model to the LGMS for measurement. Our method is helpful to solve the problems of interclass similarity and intraclass difference widely existing in remote sensing images.

D. Ablation Experiments

In this section, we will conduct the comprehensive ablation experiments. On one hand, we perform comparative experiments on ES and LGMS to analyze which one has more advantages. On the other hand, we will investigate the effectiveness of AMAM and LGIM components. We use FEM to represent the FEM in Fig. 3.

1) FEM (ES): We only consider conducting the experiments on ES. We follow the Hilbert order to flatten the feature maps of each channel in the last layer of FEM into a vector with a length of 4×4, and then concatenate the vectors of 64 channels to form a feature vector with a length of $64 \times 4 \times 4$ representing the features of the image. In the support set, we calculate the average value of all the feature vectors for each category as a prototype to represent the features of this category. Finally, Euclidean distance is used to measure the distance between the feature vectors of the query set and the prototype.

2) FEM (LGMS): We only consider conducting the experiments on LGMS. Like FEM (ES), we also flatten the feature map into a 4×4 feature vector. Chen et al. [84] proposed a method for constructing Lie groups based on the covariance matrix between different channels. Hence, we follow this idea to generate a 64×64 covariance matrix for the feature map of each image, and the covariance matrix is on LGMS. In addition, in the support set, we calculate the average value of all the feature vectors for each category of each channel and use these average value vectors from different channels to construct a covariance matrix which can be regarded as a prototype to represent this category. We use (7) to measure the distance between the covariance matrices of the features in the query set and the prototype.

3) FEM + AMAM: We add the AMAM on the basis of FEM and map its parameters to LGMS. The result is an upper triangular matrix, and we flatten the matrix into a vector. The definition of the prototype and the measurement of the distance are consistent with 1).

4) FEM + AMAM + LGIM (Ours): This is our proposed method, which adds LGIM to 3) to verify the effectiveness of the LGIM.

The experimental results are shown in Table V. We can see that the CAs of FEM (LGMS) exceed FEM (ES). The former maps the features to LGMS for measurement, while the latter directly measures the distance on ES with feature vectors. Combined with [63] and [84], we speculate that this is because the features on LGMS have stronger representation capabilities than those on ES, so that the model has a better performance. The CAs of FEM + AMAM are higher than those of FEM (ES) and FEM (LGMS), which indicates that the AMAM component is effective. The component can model the observation features and the space state of the features and describe the spatial correlation between local features of remote sensing images. In addition, we added the LGIM components on the basis of FEM + AMAM and established the FEM + AMAM + LGIM (ours) method, which significantly improves the CAs and outperforms all the other methods. LGIM is a prototype defined based on the LGMS, which can effectively preserve the geometric structure of the manifold, reduce the loss of information, and improve the performance of the model.

E. Influence of Different Flattening Orders of Local Features on CAs

In Section I, we have introduced the influence of different scanning sequences on spatial relationships of local image patches. Among them, we flattened the image patches according to three different scanning sequences, i.e., scanning by row sequence, scanning by column sequence, and scanning by Hilbert sequence. Through the visualization results, we can see that the image patches flattened according to the Hilbert sequence can better maintain the local adjacent relationship of the image. In this section, we intend to investigate the influence of different flattening orders of local features on CAs. We flatten the feature maps in three different orders (i.e., row order, column order, and Hilbert curve order), Then, we carry out RSSC experiments on three public datasets according to the method proposed in Section III. We name the three methods HFS-LGMS (row), HFS-LGMS (column), and HFS-LGMS (ours). Finally, we compare the CAs of these methods.

Table VI shows the experimental results with three methods based on different flattening orders of local features. We can see that HFS-LGMS (ours) achieves the highest CAs on all the three datasets. In addition, compared with HFS-LGMS (ours), HFS-LGMS (row) and HFS-LGMS (column) have no obvious advantages. On all the datasets, the CAs' difference between HFS-LGMS (row) and HFS-LGMS (column) is within 1%. For the NWPU dataset, the performance of HFS-LGMS (row)

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TABLE VI
FIVE-WAY FSL CAS (%) FOR RSSC ON THREE DATASETS WITH THREE METHODS BASED ON DIFFERENT FLATTENING ORDERS OF LOCAL FEATURES

Settings	NWPU		AID		UCM	
	One-shot	Five-shot	One-shot	Five-shot	One-shot	Five-shot
HFS-LGMS (row)	76.25±1.27	86.4± 0.46	64.09±0.72	76.88±0.81	60.31±0.54	79±0.13
HFS-LGMS (column)	75.47±0.38	85.81±0.23	65.07±0.65	76.91±0.16	61±0.51	78.91±0.82
HFS-LGMS (ours)	**76.81±0.18**	**87.14±0.35**	**65.96±0.15**	**77.86±0.68**	**63.05±0.78**	**79.68±0.81**

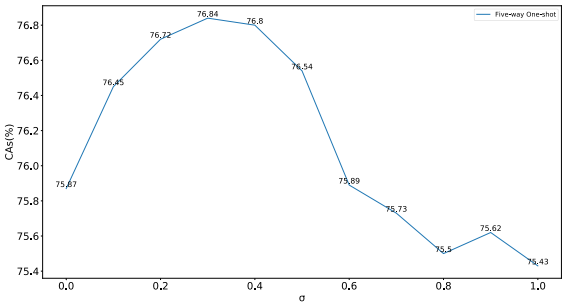

Fig. 4. Five-way one-shot CAs (%) on the NWPU dataset when σ is changed from 0 to 1.

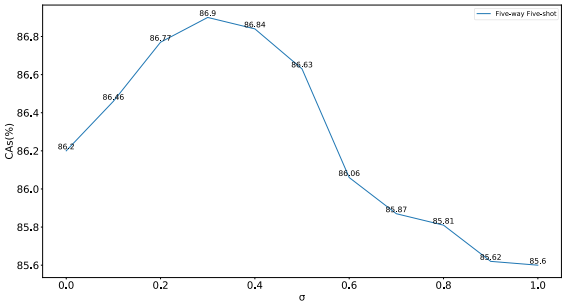

Fig. 5. Five-way five-shot CAs (%) on the NWPU dataset when σ is changed from 0 to 1.

outperforms the performance of HFS-LGMS (column). For the AID dataset, the performance of HFS-LGMS (column) outperforms the performance of HFS-LGMS (row). However, the results are slightly different on the UCM dataset, and the CAs of HFS-LGMS (row) are lower than that of HFS-LGMS (column) in the case of five-way one-shot. But in the case of five-way five-shot, the results are opposite. Hence, we can conclude that the strategy of flattening features along the Hilbert curve is beneficial to improve the CAs of RSSC.

F. Determination of Optimal Hyperparameter

In (11), σ is a hyperparameter. To get the optimal hyperparameter, we set σ to a group of changing values, i.e., [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]. In the five-way one-shot group and in the five-way five-shot group, we conduct repeated experiments on the NWPU dataset under different σ values to investigate the setting of σ when the model achieved optimal performance.

The experimental results are shown in Figs. 4 and 5. Fig. 4 displays five-way one-shot CAs on the NWPU dataset when σ is changed from 0 to 1. We find that the model achieves the highest CA when σ is set to 0.3, where CA is 76.84%. Fig. 5 displays five-way five-shot CAs on the NWPU dataset when σ is changed from 0 to 1. It is obvious that the model achieved optimal performance when σ is set to 0.3, where CA reaches 86.9%. Hence, in all the comparative experiments, we set σ to 0.3.

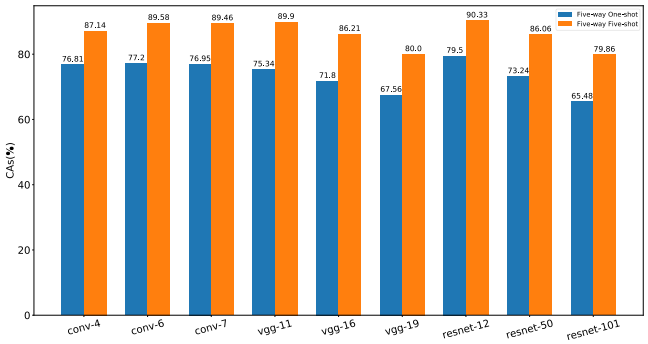

Fig. 6. CAs (%) on the NWPU dataset under different backbone networks.

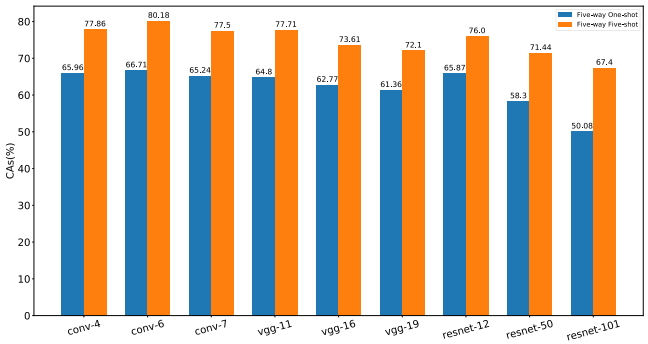

Fig. 7. CAs (%) on the AID dataset under different backbone networks.

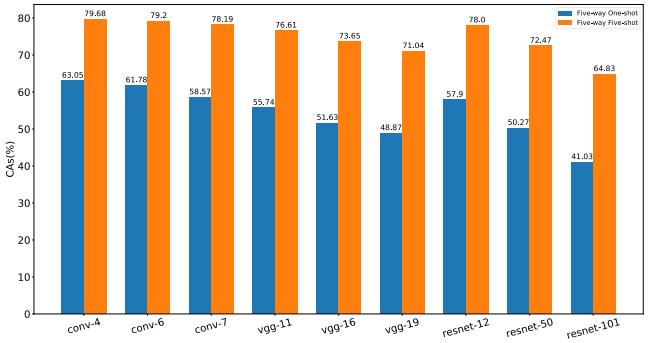

Fig. 8. CAs (%) on the UCM dataset under different backbone networks.

G. Impact of Different Backbone Networks

To further demonstrate the performance of our method, we conduct experiments on different backbone networks with three public datasets. We selected three different backbone networks, i.e., lightweight shallow CNNs (conv-4, conv-6, and conv-7), VGG-based networks (vgg-11, vgg-16, and vgg-19), and resnet-based networks (resnet-12, resnet-50, and resnet-101). The experimental results are shown in Figs. 6–8.

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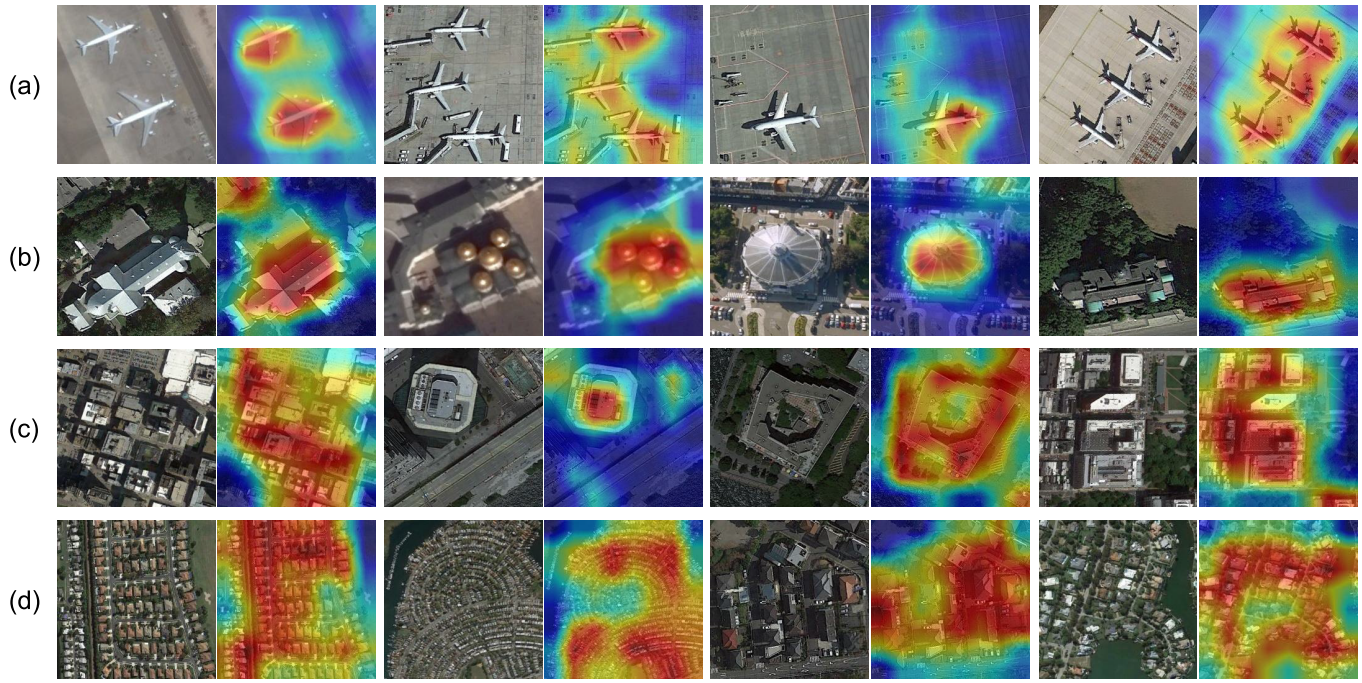

Fig. 9. Comparison between the original image and class activation map on the five-way five-shot task. (a) Airplane. (b) Church. (c) Commercial area. (d) Dense residential.

It can be seen from Fig. 6, for shallow CNNs, our method achieves the best performance on conv-6. For the VGG-based networks, vgg-11 gets the highest CA. For the resnet-based networks, resnet-12 gets the highest CA. And the performance of our method on vgg-11 and resnet-12 exceeds conv-6. This result indicates that our method can also achieve satisfied results on deep backbone networks on the NWPU dataset, which may be because the NWPU dataset contains richer samples. From Fig. 7, we can see that our method achieves the highest CA on conv-6. Fig. 8 shows that our method achieves the highest CA on conv-4. The experimental results on three datasets show a common rule, that is, for VGG-based networks and resnet-based networks, with the number of network layers increasing, the performance of our method becomes worse. For FSL, the number of samples that can be used each time during the training process is limited. Hence, as the number of layers of the network increases, the phenomenon of overfitting becomes more and more serious, which eventually leads to the degradation of model performance.

H. Analysis of Class Activation Map

To show the feature extraction ability more intuitively of our method, we use Grad-CAM [85] to draw class activation map. We select four common remote sensing scenes (airplane, church, commercial area, and dense residential) from the NWPU dataset and randomly select four images for visual analysis in each scene on the five-way five-shot task model.

The results are shown in Fig. 9. The class activation map in Fig. 9 is rendered with a gradient color from blue to red. The redder the color, the greater the contribution of the ground objects contained in the region to scene classification. On the contrary, the bluer the color, the less the contribution of the region. By observing Fig. 9, we can see that our proposed method can accurately capture the target areas which contribute greatly to scene classification. For example, for airplane, our method can easily distinguish the airplane from the ground background. For the church, the spire and dome of the building are very representative, and these obvious features show dark red in the heat map. For commercial area and dense residential, the target area is difficult to be completely identified due to its large scope. However, for most images, this information is sufficient to discriminate different scenes.

TABLE VII

COMPARISONS OF THE RUNNING TIME OF HFS-LGMS IN THE CASE OF FIVE-WAY ONE-SHOT UNDER THE NWPU DATASET

Methods	Train/Epoch(s)	Test/Task(s)
MN [75]	32.28	1.23
Meta-SGD [76]	39.66	1.28
RN [77]	40.27	1.38
PN [46]	38.96	1.27
DLA-MatchNet	48.67	1.58
MAML	37.79	1.26
SCL-MLNet [47]	52.78	1.67
TSC [33]	56.43	1.73
CovaMNet [63]	58.5	1.77
HFS-LGMS (ours)	60.74	1.84

I. Running Time Analysis

To investigate the efficiency and complexity of HFS-LGMS, we perform comparative experiments on the NWPU dataset and compare the training time of each epoch and the test time of each five-way one-shot task of our method with some classical FSL methods. The experimental results are reported in Table VII. We can observe that HFS-LGMS consumes more time in the training and testing stages than other methods. The training time of each epoch is 2.24 s longer than that of TSC [18], and the test time of each five-way one-shot task

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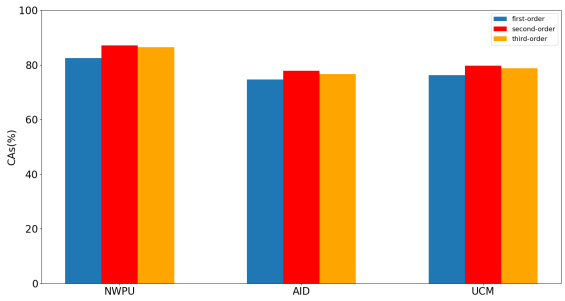

Fig. 10. Five-way five-shot CAs (%) on the NWPU dataset with different orders of Hilbert curves.

is 0.07 s longer than that of TSC [18]. It is obvious that the increased time cost of the test task can be almost negligible. Hence, the efficiency and complexity of our proposed method are acceptable.

J. Impact of Different Orders of Hilbert Curves

To study the impact of different orders of the Hilbert curves on CAs, we conduct comparative experiments on the NWPU dataset. In the case of five-way five-shot, we flatten the local features in the order of first-order, second-order and third-order Hilbert curves. The experimental results in Fig. 10 indicate that modeling based on second-order Hilbert curve reaches the best performance. Hence, we follow the second-order Hilbert curve to flatten the feature map in all the experiments in this article.

V. Conclusion

This article proposes an RSSC framework based on FSL called HFS-LGMS. Considering the influence of the arrangement order of high-level features of remote sensing images for RSSC with FSL methods, we flatten local features along the Hilbert curve to better maintain the spatial adjacent relationship of remote sensing image features. Then, we introduce the AMAM to model the observation features and the space state of the features and describe the spatial correlation between local features of remote sensing images. The parameters of AMAM are embedded into LGMS, and the prototype of the features on Lie group space is defined by LGIM. Finally, RSSC is implemented in Lie group metric space based on geodesic distance. The experimental results on three public datasets indicate that our method is superior to most state-of-the-art methods.

However, there is still a problem that our proposed method can only be trained on labeled samples. In the field of remote sensing, it is very expensive to obtain labeled samples, which not only takes a lot of time but also requires rich experience and professional knowledge. Therefore, in our future research, we will consider further expanding our approach to combine semi-supervised or self-supervised learning so that unlabeled samples can be leveraged to reduce the reliance on huge number of labeled samples.

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